In [ ]:
```
print("Hello, Machine learning!")
```

```
Hello, Machine learning!
```

In [ ]:
```
print("I am Alhanouf and I am learning Python using Google Colab")
```

```
I am Alhanouf and I am learning Python using Google Colab
```

In [1]:
```
name = "Alhanouf"
age = 26
height = 151
is_student = True
print(name, age, height, is_student)
print(type(name), type(age), type(height), type(is_student))
```

```
Alhanouf 26 151 True
<class 'str'> <class 'int'> <class 'int'> <class 'bool'>
```

In [ ]:
```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
print("Hello", name)
print("next year you will be", age + 1)
```

```
Enter your name: hanouf
Enter your age: 26
Hello hanouf
next year you will be 27
```

In [2]:
```
a = 5
b = 5
print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Floor division:", a // b)
print("Remainder:", a % b)
print("Power:", a ** b)
```

```
Addition: 10
Subtraction: 0
Multiplication: 25
Division: 1.0
Floor division: 1
Remainder: 0
Power: 3125
```

In [ ]:
```
mark = float(input("Enter your mark: "))
if mark >= 90:
  print("A")
elif mark >= 80:
  print("B")
elif mark >= 70:
  print("C")
elif mark >= 60:
    print("D")
```

```
else:
    print("Grade: F")
```

```
Enter your mark: 89
B
```

In [11]:
```
temp = [24.4, 25.1, 26.3, 27.0, 28.8]
print("Number of observations:", len(temp))
print("Average temperature:", sum(temp) / len(temp))
print("Minimum temperature:", min(temp))
print("Maximum temperature:", max(temp))
```

```
Number of observations: 5
Average temperature: 26.32
Minimum temperature: 24.4
Maximum temperature: 28.8
```

In [ ]:
```
marks = [65, 82, 70, 60, 90]
print("Marks:", marks)
print("First mark:", marks[0])
print("Third mark:", marks[2])
print("Number of marks:", len(marks))
print("Total:", sum(marks))
print("Highest:", max(marks))
print("Lowest:", min(marks))
print("Average:", sum(marks) / len(marks))
```

```
Marks: [65, 82, 70, 60, 90]
First mark: 65
Third mark: 70
Number of marks: 5
Total: 367
Highest: 90
Lowest: 60
Average: 73.4
```

In [14]:
```
study_hours = [1, 2, 3, 4, 5, 6, 7]
marks = [50, 60, 70, 75, 80, 83, 90]
for i in range(len(study_hours)):
 print("Study hours:", study_hours[i], "Marks:", marks[i])
```

```
Study hours: 1 Marks: 50
Study hours: 2 Marks: 60
Study hours: 3 Marks: 70
Study hours: 4 Marks: 75
Study hours: 5 Marks: 80
Study hours: 6 Marks: 83
Study hours: 7 Marks: 90
```

In [15]:
```
study_hours = [1, 2, 3, 4, 5, 6, 7]
for mark in marks:
  if mark >= 60:
    print(mark, "Pass")
  else:
    print(mark, "Fail")
```

```
50 Fail
60 Pass
70 Pass
75 Pass
80 Pass
83 Pass
90 Pass
```

In [16]:
```python
print("Number of samples:", len(marks))
average_hours = sum(study_hours) / len(study_hours)
print("Average study hours:", average_hours)
average_mark = sum(marks) / len(marks)
print("Average mark:", average_mark)
print("Highest mark:", max(marks))
print("Lowest mark:", min(marks))
print("\nStudy Hours / Mark:")
for i in range(len(marks)):
    print(study_hours[i], "hours:", marks[i])
print("\nPass/Fail:")
for mark in marks:
    if mark >= 60:
        print(mark,":Pass")
    else:
        print(mark,":Fail")
passes = 0
for mark in marks:
    if mark >= 60:
        passes += 1
print("\nNumber of students who passed:", passes)
```

```
Number of samples: 7
Average study hours: 4.0
Average mark: 72.57142857142857
Highest mark: 90
Lowest mark: 50

Study Hours / Mark:
1 hours: 50
2 hours: 60
3 hours: 70
4 hours: 75
5 hours: 80
6 hours: 83
7 hours: 90

Pass/Fail:
50 :Fail
60 :Pass
70 :Pass
75 :Pass
80 :Pass
83 :Pass
90 :Pass

Number of students who passed: 6
```